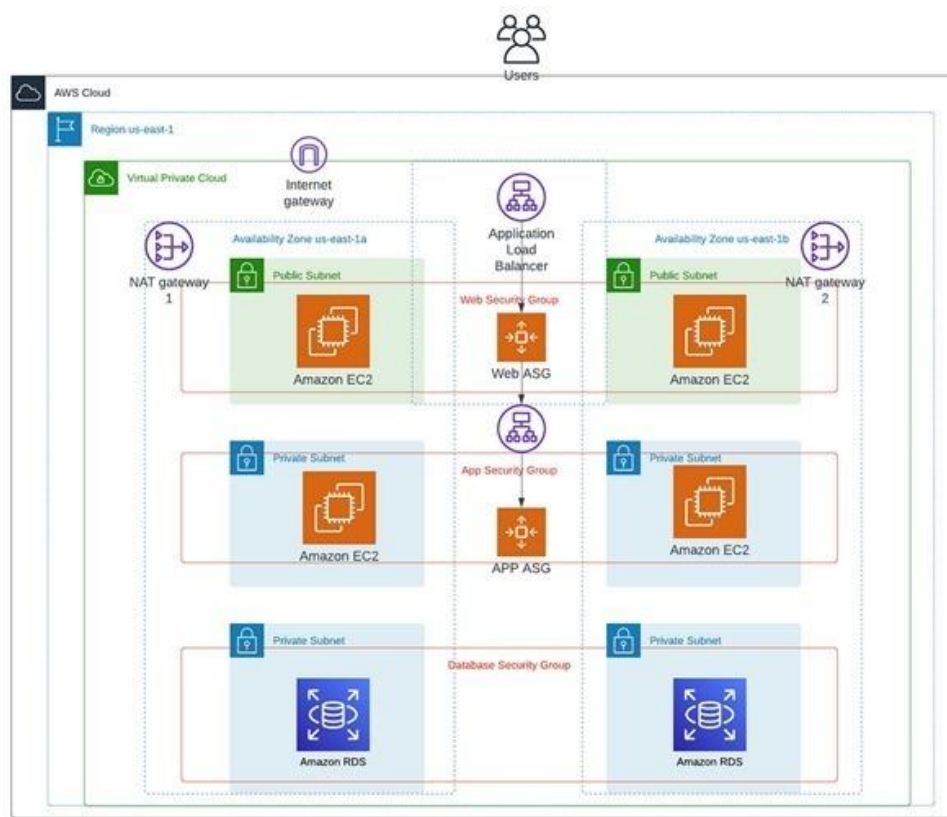
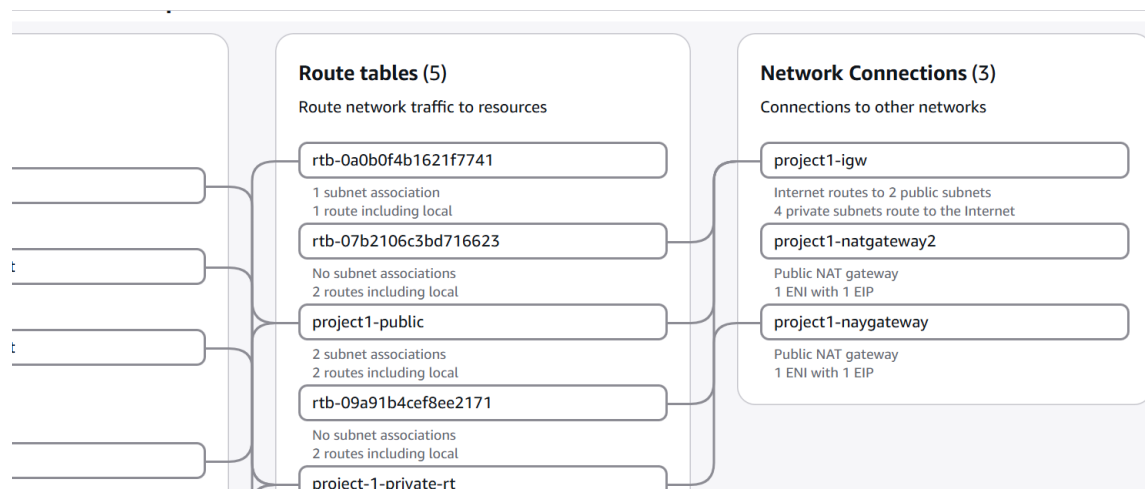


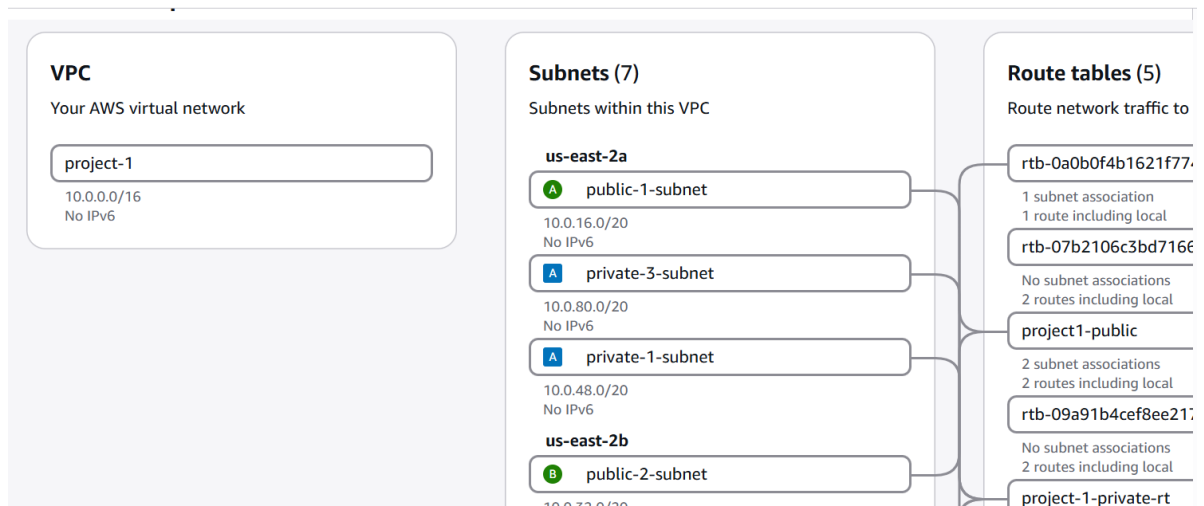
Project-1

Design and Implementation of a Highly Available Three-Tier Web Application on AWS



VPC:





INTERNET GATEWAY:

igw-04d21877023ffab88 / project1-igw Actions

Details [Info](#)

Internet gateway ID  igw-04d21877023ffab88	State  Attached	VPC ID vpc-092466655fcba3772 / project-1	Owner  941718377056
--	---	--	---

Tags (1) Manage tags

Key	Value
Name	project1-igw

NAT Gateway (for two public servers):

NAT gateways (2) [Info](#) Actions Create NAT gateway

	Name	NAT gateway ID	Connectivity...	State	State message	Availabi
☐	project1-natgateway2	nat-1004d2cef864019d8	Public	 Available	-	Regiona
☐	project1-naygateway	nat-1241cea8a52761f47	Public	 Available	-	Regiona

Select a NAT gateway

Target groups:

EC2 > Target groups > project-1

Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups
Trust Stores

▼ Auto Scaling
Auto Scaling Groups

Settings

Registered targets (4) Info [Anomaly mitigation: Not applicable](#) [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health status details
☐	i-0e800c112639dcf13	private-2	80	us-east-2b (us-east-2)	Healthy	-
☐	i-0e9ced87bf36e96b2	private-1	80	us-east-2a (us-east-2)	Healthy	-
☐	i-081bd47d93788b244	public-2	80	us-east-2b (us-east-2)	Healthy	-
☐	i-0908c36d8217bd351	public-1	80	us-east-2a (us-east-2)	Healthy	-

Load balance:

EC2 > Load balancers > project1-lb

Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups
Trust Stores

▼ Auto Scaling
Auto Scaling Groups

Settings

▼ Details

Load balancer type Application	Status Active	VPC vpc-092466655fcb3772	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z3AADJGX6KTTL2	Availability Zones subnet-0de26ab8cf915b6d2 us-east-2a (use2-az1) subnet-02fe9a3096208dbaa us-east-2b (use2-az2)	Date created September 17, 2026, 21:41 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-2:941718377056:loadbalancer/app/project1-lb/c1c52768debf6659		DNS name Info project1-lb-1928990012.us-east-2.elb.amazonaws.com (A Record)	

AMI:

EC2 > AMIs

Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups

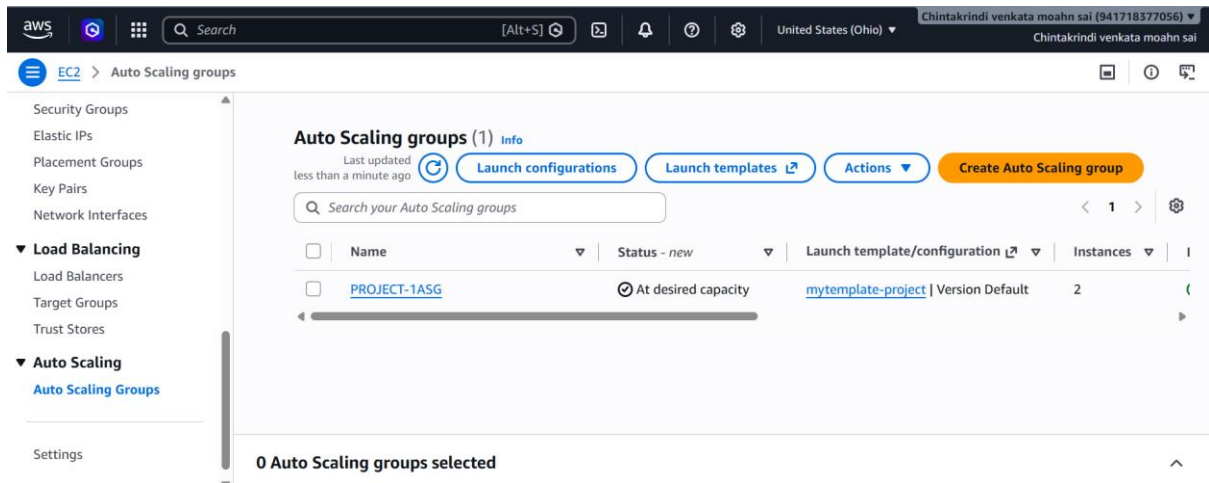
Amazon Machine Images (AMIs) (3) Info [Recycle Bin](#) [EC2 Image Builder](#) [Actions](#) [Launch instance from AMI](#)

Owned by me Filter AMI by attribute or tag

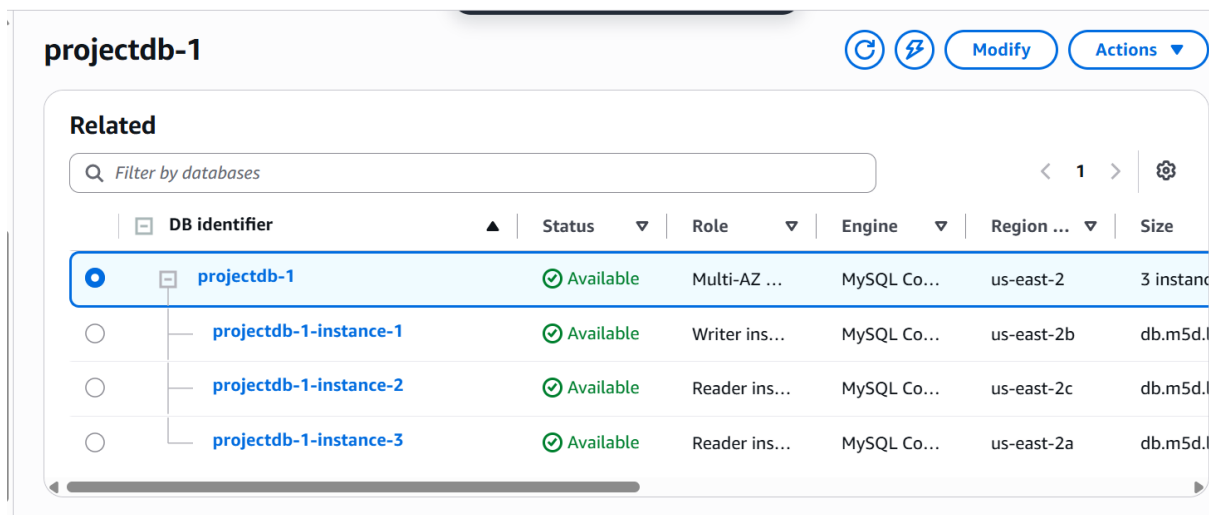
☐	Name	AMI name	AMI ID	Source	Own
☐	public-image		ami-0b2e4d811cc3d9ba2	941718377056/public-image	9417
☐	private-image		ami-0a1542f35cab8410b	941718377056/private-image	9417

Select an AMI

Auto-scaling:



RDS:



HERE I USED MULTI-AZ. FOR THAT, WE NEED 3 SUBNETS, SO I CREATED ONE MORE SUBNET

1.PRIVATE-1 :

From the public, we need to connect to the private server:

```
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| PROJECT |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.002 sec)

MySQL [(none)]> use PROJECT;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [PROJECT]> SELECT * FROM employees;
+----+-----+-----+
| id | name  | salary |
+----+-----+-----+
| 1  | Mohan | 30000  |
| 2  | Ravi  | 35000  |
| 3  | Kiran | 40000  |
+----+-----+-----+
3 rows in set (0.001 sec)

MySQL [PROJECT]> |
```


Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;

```
+-----+
| Database |
+-----+
| PROJECT |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
```

5 rows in set (0.002 sec)

MySQL [(none)]> use PROJECT;

Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed

MySQL [PROJECT]> SELECT * FROM employees;

```
+-----+-----+-----+
| id | name | salary |
+-----+-----+-----+
| 1 | Mohan | 30000 |
| 2 | Ravi | 35000 |
| 3 | Kiran | 40000 |
+-----+-----+-----+
```

3 rows in set (0.001 sec)

MySQL [PROJECT]> |

2.connecting to private server 2:

PS C:\Users\chmoh\onedrive\desktop> ssh -i "project-1.pem" ec2-user@3.142.246.189

```
#
#####
\#####\
#####|
\#/\
V--> https://aws.amazon.com/linux/amazon-linux-2023
```

Last login: Fri Sep 18 10:52:57 2026 from 152.57.228.218

[ec2-user@ip-10-0-32-98 ~]\$ sudo -i

[root@ip-10-0-32-98 ~]# yum install nginx -y

Last metadata expiration check: 5:46:00 ago on Fri Sep 18 08:20:44 2026.

Package nginx-1:1.30.4-1.amzn2023.0.1.x86_64 is already installed.

Dependencies resolved.

Nothing to do.

Complete!

[root@ip-10-0-32-98 ~]# systemctl start nginx

[root@ip-10-0-32-98 ~]# ssh -i "project-1.pem" ec2-user@10.0.67.220

A newer release of "Amazon Linux" is available.

Version 2023.12.20260917:

Run "/usr/bin/dnf check-release-update" for full release and version update info

```
#
#####
\#####\
#####|
\#/\
V--> https://aws.amazon.com/linux/amazon-linux-2023
```

```

      _.._
     /_/_/_/_/
    /m/'

Last login: Fri Sep 18 10:07:39 2026 from 10.0.32.98
[ec2-user@ip-10-0-67-220 ~]$ sudo -i
[root@ip-10-0-67-220 ~]# yum install nginx -y
Last metadata expiration check: 6:18:12 ago on Fri Sep 18 07:49:44 2026.
Package nginx-1:1.30.4-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-10-0-67-220 ~]# systemctl start nginx
[root@ip-10-0-67-220 ~]# yum install mariadb105-server -y
Last metadata expiration check: 6:18:35 ago on Fri Sep 18 07:49:44 2026.
Package mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-10-0-67-220 ~]# systemctl start mariadb
[root@ip-10-0-67-220 ~]# mysql -h projectdb-1.cluster-ro-clc8c6kmggsp.us-east-2.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 539
Server version: 8.4.9 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+

```

```

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| PROJECT |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.002 sec)

MySQL [(none)]> use PROJECT;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [PROJECT]> SELECT * FROM employees;
+-----+-----+-----+
| id | name | salary |
+-----+-----+-----+
| 1 | Mohan | 30000 |
| 2 | Ravi | 35000 |
| 3 | Kiran | 40000 |
+-----+-----+-----+
3 rows in set (0.001 sec)

MySQL [PROJECT]> |

```

THE DATA IN PRIVATE SERVER, NEED TO SEE IN PUBLIC SERVER:


```

MySQL [(none)]> show databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'show databases' at line 1
MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| PROJECT |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.001 sec)

MySQL [(none)]> use PROJECT;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [PROJECT]> SELECT * FROM employees;
+-----+
| id | name | salary |
+-----+
| 1 | Mohan | 30000 |
| 2 | Ravi | 35000 |
| 3 | Kiran | 40000 |
+-----+
3 rows in set (0.000 sec)

MySQL [PROJECT]>

```

When the server is overloaded, the new servers are automatically created by autoscaling,

The screenshot shows the AWS Management Console interface for EC2 Instances. On the left, there is a navigation menu with options like Dashboard, AWS Global View, Events, and a list of instance types (Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations). The main content area shows a list of 6 instances, all in a 'Running' state. Each instance has a unique ID, a name (e.g., public-1, private-2), and is located in a specific availability zone (us-east-2a or us-east-2b). A green banner at the top of the console indicates a successful action: 'Deleting the instance or carrying out other actions related to this AMI.' Below the instance list, there is a section for 'All instances'.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
public-1	i-0908c36d8217bd351	Running	t3.micro	3/3 checks passed	us-east-2a
	i-08bb2a5e47418d917	Running	t3.micro	3/3 checks passed	us-east-2a
public-2	i-081bd47d93788b244	Running	t3.micro	3/3 checks passed	us-east-2b
private-1	i-0e9ced87bf36e96b2	Running	t3.micro	3/3 checks passed	us-east-2a
private-2	i-0e800c112639dcf13	Running	t3.micro	3/3 checks passed	us-east-2b
	i-087abe7ace23ae858	Running	t3.micro	3/3 checks passed	us-east-2b